

Music Genre Classification using Deep Learning and Transfer Learning

Project Report

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Neural Networks and Deep Learning

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1 Introduction

In this project, we work on automatic music genre classification using machine learning and deep learning. Identifying music genres from audio files is an interesting problem because audio signals are complex and genre boundaries are often subjective.

Using the Free Music Archive dataset (`fma_small`), our goal is to classify 30-second audio clips into 8 different genres. We start by analyzing simple audio features, build baseline statistical models, and then test Convolutional Neural Networks (CNNs) as well as transfer learning techniques using Google’s VGGish model.

2 Problem Definition

This project addresses automatic music genre classification, a classic task in Music Information Retrieval (MIR) at the intersection of audio signal processing and deep learning. The main goal is to predict the musical genre of a 29-second raw audio clip among 8 possible categories.

This topic was chosen for two main reasons:

1. It allows us to explore multiple ways of representing audio signals for deep learning models (such as mel-spectrograms and pre-trained embeddings), which is a rich and non-trivial task.
2. Music genre classification has an interesting difficulty: boundaries between genres are often blurry and subjective (for example, a song can reasonably sit between Pop and Rock). This makes it a great benchmark to observe the real limitations of classification models beyond simple accuracy scores.

The problem is formulated as a supervised multi-class classification problem (8 balanced classes) evaluated on a standardized train/validation/test split.

3 Data Collection

We use the Free Music Archive (FMA) dataset, specifically the `fma_small` subset. This open dataset is widely used in research for music genre classification. It contains 8,000 tracks evenly distributed across 8 genres (1,000 tracks per genre): Hip-Hop, Pop, Folk, Experimental, Rock, International, Electronic, and Instrumental. Each audio clip lasts 30 seconds.

The data consists of two parts:

- A metadata file (`tracks.csv`) containing track details (genre, artist, album, release date, official split, etc.).
- Raw audio files in `.mp3` format, organized in subfolders by track ID.

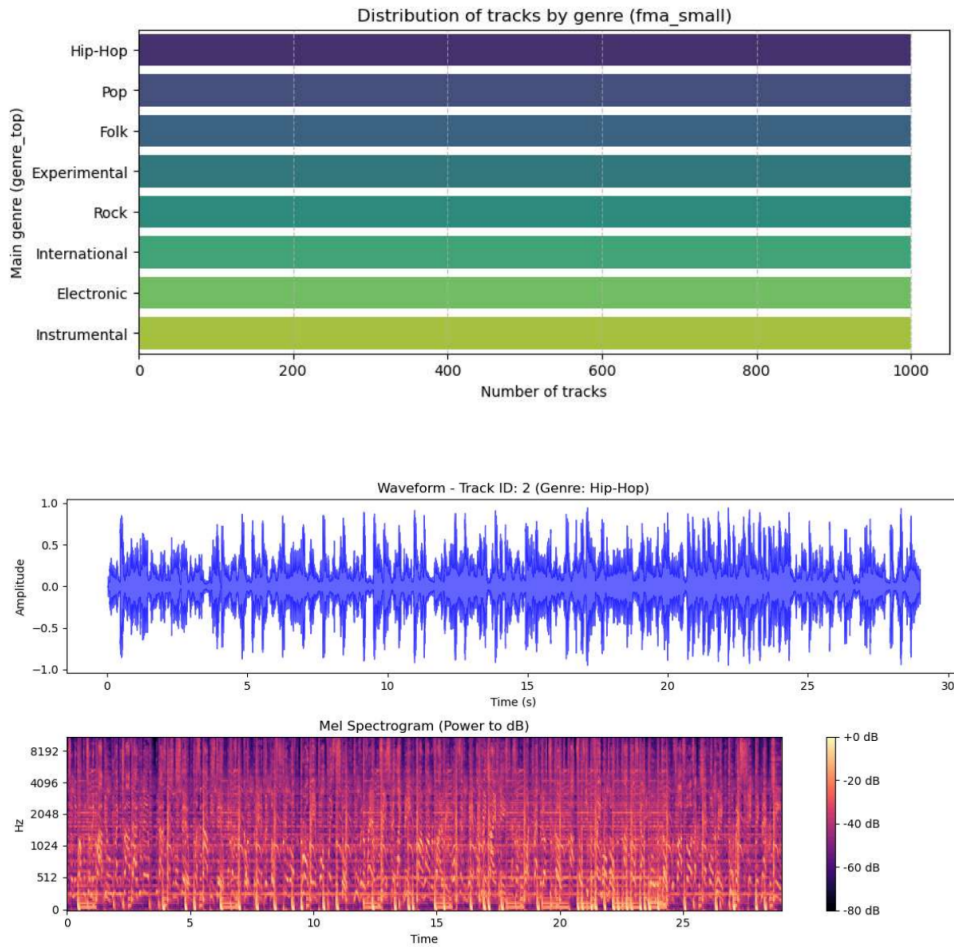
We use the official FMA train/validation/test split (80% / 10% / 10%, corresponding to 6,400 / 800 / 800 tracks). This guarantees a standardized evaluation comparable with existing literature and prevents data leakage between tracks from the same artist.

4 Exploratory Data Analysis (EDA)

4.1 Genre Distribution and Data Quality

The genre distribution is perfectly balanced with 1,000 tracks per class. This eliminates class bias: the trained model will not favor a majority class, and classification accuracy serves as a reliable performance metric.

No missing values were detected in key metadata columns (duration, genre, artist, album, creation date). The average track length in the original archive is ~ 229 seconds (standard deviation ~ 98 seconds), confirming that our 29-second fixed extract provides a representative sample of each song’s content.



4.2 Audio Signal Inspection

Inspecting a Hip-Hop track highlights the two main audio representations used in this project:

- **Waveform:** Audio amplitude over time.
- **Mel-Spectrogram:** Distribution of frequency energy over time in decibels (dB).

The mel-spectrogram is the chosen representation for deep learning models because it captures both temporal and spectral information, unlike raw waveforms which are difficult for standard convolutional networks to process directly.

4.3 Classical Acoustic Feature Extraction

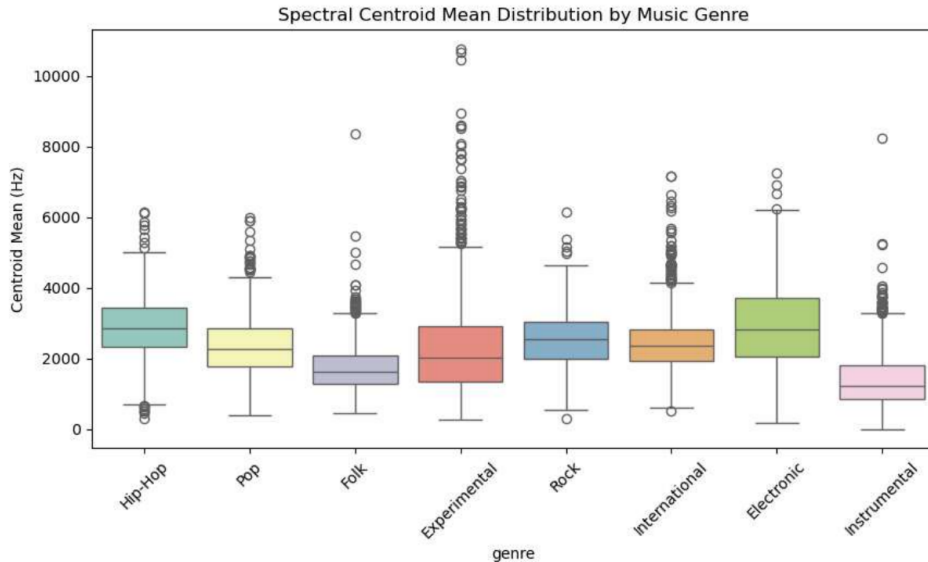
In addition to deep learning, we extracted classical acoustic features (20 MFCC coefficients, chroma, spectral contrast, spectral centroid, spectral bandwidth, spectral rolloff, zero-crossing rate, RMS energy, and tempo). This resulted in 7,994 usable tracks out of 8,000 (99.9% success rate, with minor failures due to corrupted files distributed evenly across genres).

4.4 Spectral Centroid Analysis by Genre

The spectral centroid measures the frequency "brightness" of a sound and shows noticeable differences across genres:

- **Electronic** and **Hip-Hop** display the highest spectral centroids, consistent with synthetic sounds rich in high frequencies.
- **Folk** and **Instrumental** show lower centroids due to the prevalence of acoustic instruments.
- **Experimental** shows the highest variance, reflecting the intrinsic diversity of this genre.

However, these distributions overlap significantly, demonstrating that a single feature is not enough and justifying multi-dimensional feature sets.



4.5 Descriptor Correlations and Feature Selection

The correlation matrix reveals strong redundancy between centroid, bandwidth, and rolloff ($r > 0.9$), as all three measure frequency energy concentration. In contrast, RMS energy is largely independent of frequency descriptors ($r < 0.25$), making it a useful complementary feature.

To select the best features, we first removed irrelevant metadata, uninformative high-order MFCCs, and low-variance columns, reducing the feature space from 93 to 74 columns. We then applied an ANOVA F -test to evaluate the discriminative power of each remaining feature across the 8 genres. Spectral descriptors (rolloff, contrast, centroid, bandwidth) and low-order MFCCs were found to be the most discriminative. Retaining the top 30 descriptors yielded a final vector of 72 features (after excluding track IDs and labels), reducing overfitting while preserving essential signal information.

5 Baseline Models

We trained three baseline models on the 72 standardized tabular features (with scaling parameters fitted strictly on the training set to avoid data leakage):

Table 1: Baseline Performance Comparison on Tabular Features

Model	Accuracy (Val / Test)
Logistic Regression	46.6%
Random Forest (200 trees)	52.6%
Simple MLP ($72 \rightarrow 128 \rightarrow 64 \rightarrow 8$)	$\sim 51.6\%$

Logistic Regression showed heterogeneous performance across genres: Folk ($F_1 = 0.66$) and Rock ($F_1 = 0.60$) were well recognized, whereas Pop was almost always misclassified ($F_1 = 0.10$). Random Forest improved accuracy to 52.6% thanks to its ability to model non-linear feature interactions, though Pop remained weak ($F_1 = 0.18$). The MLP achieved similar results ($\sim 51.6\%$).

The convergence of linear, tree-based, and neural models around a ~ 50 – 53% performance plateau suggests that the limitation lies in the summary nature of tabular features rather than the learning algorithms.

6 Deep Learning Architecture

6.1 Framework Choice and Development Environment

For the development of our deep learning architectures, we chose PyTorch instead of TensorFlow. This choice was driven by initial technical constraints: hardware and environment compatibility issues on the local machine, as well as difficulties when loading and hosting the FMA dataset on Google Colab, made TensorFlow complicated to use.

Switching to PyTorch proved to be very practical. Beyond our previous experience with the tool, PyTorch offers great flexibility for dynamic tensor manipulation, seamless audio data loading with `DataLoader`, and easy integration of pre-trained models for transfer learning.

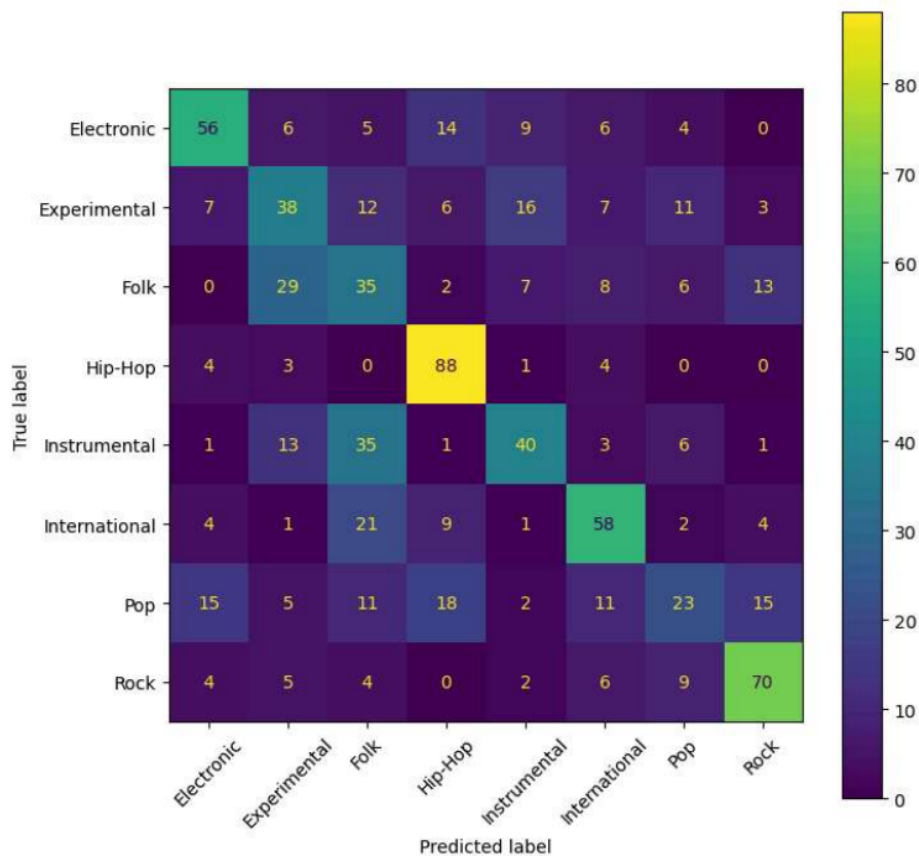
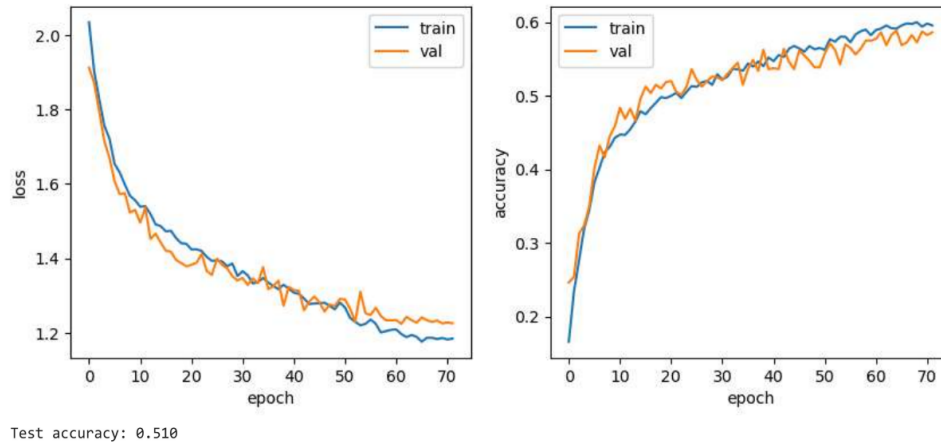
6.2 CNN on Mel-Spectrograms (v1)

To preserve spatio-temporal structure, we developed a Convolutional Neural Network operating on 2D mel-spectrograms:

- 3 convolutional blocks ($16 \rightarrow 32 \rightarrow 64$ channels, 3×3 kernels) with MaxPooling.
- Global Average Pooling to handle variable duration.
- A dense classification head ($64 \rightarrow 32 \rightarrow 8$) with Dropout (0.3).

Using early stopping and learning rate decay, training stopped at epoch 72. The model achieved 51.0% test accuracy, comparable to tabular baselines.

The confusion matrix showed distinct genre behaviors: Hip-Hop had high recall (88/100) due to its unique rhythmic signature. Folk and Instrumental were frequently confused with each other. Pop remained the hardest class (23/100 correct predictions), spreading across almost all other categories due to its broad, commercial acoustic diversity.



7 Optimization and Improvements

To break through the $\sim 51\%$ accuracy ceiling, we implemented several optimizations:

1. **Data Augmentation (SpecAugment):** Random frequency and time masking applied to spectrograms, forcing the model to learn robust global patterns rather than local noise.
2. **Deeper Architecture (CNN v2):** Added a 4th convolutional block (up to 128 channels) and Batch Normalization after each convolution to stabilize training.
3. **Regularization:** Added weight decay ($L_2 = 10^{-4}$) alongside Dropout (0.4).

CNN v2 reached a test accuracy of 53.6% (+2.6% over v1) and converged significantly faster (validation accuracy exceeded 60% by epoch 19).

Evaluating top-2 accuracy yielded **72.3%** (compared to 53.6% top-1 accuracy). This +18.7% gap indicates that model errors are not random; the network usually identifies the correct stylistic family but hesitates between closely related genres (e.g., Folk vs. Instrumental, Pop vs. Rock).

7.1 Transfer Learning with VGGish

Our final approach leveraged Google’s VGGish, a deep CNN pre-trained on AudioSet (~ 2 million audio clips). VGGish converts each 1-second audio segment into a 128-dimensional semantic embedding.

For each track, we concatenated the mean and maximum of its patch embeddings into a 256-dimensional representation, capturing both overall content and salient audio peaks. A lightweight MLP classifier ($256 \rightarrow 128 \rightarrow 64 \rightarrow 8$) was then trained on top of these frozen embeddings.

Training converged rapidly: validation accuracy reached 60.9% at epoch 1, and early stopping triggered at epoch 13. On the test set, VGGish reached **56.4% accuracy**, the top performance in this project.

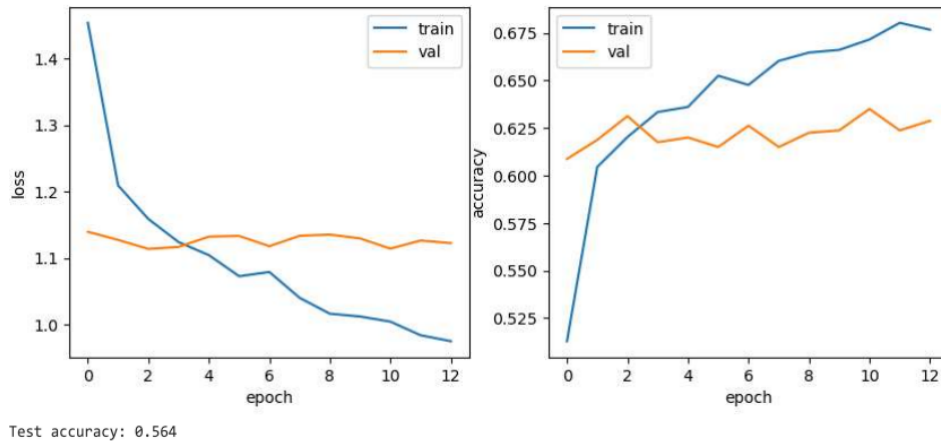


Table 2: Overall Performance Summary across Models

Model Strategy	Test Accuracy
Logistic Regression (Tabular Features)	46.6%
Random Forest (Tabular Features)	52.6%
MLP (Tabular Features)	$\sim 51.6\%$ (Val)
CNN v1 (Mel-Spectrograms, From Scratch)	51.0%
CNN v2 (SpecAugment + BatchNorm + L_2)	53.6%
VGGish Transfer Learning + MLP	56.4%

8 Critical Interpretation of Results

8.1 Understanding the 56.4% Score

An accuracy of 56.4% across 8 balanced classes (where random guessing gives 12.5%) demonstrates genuine learning. However, understanding why accuracy struggle at this level provides valuable insights into audio classification problems.

8.2 The Pop Genre as a Case Study

Across all models, Pop was consistently the hardest genre to classify (F_1 -score between 0.10 and 0.40). Because Pop is a commercial category rather than a unified acoustic style, it borrows elements from Rock, Electronic, and International music. Its lack of a distinct spectral footprint causes intrinsic overlap with other classes.

8.3 Top-1 vs. Top-2 Accuracy Gap

The gap between top-1 (53.6%) and top-2 (72.3%) accuracy confirms that misclassifications occur primarily between stylistically adjacent genres. For example, a Pop track might be predicted as Rock with 51.9% confidence, while Pop receives 34.8% confidence—a case where human listeners would also find the boundary ambiguous.

8.4 Strengths and Weaknesses of the Final Model

- **Strengths:** VGGish transfer learning offers the best performance-to-complexity ratio. Pre-trained representations generalize much better than training from scratch on limited samples (6,400 tracks), while drastically reducing training time (13 epochs vs. 72 epochs).
- **Weaknesses:** Simple mean-max pooling flattens long-term sequential structure across the 29-second clip. The final head also showed mild overfitting (`train_acc` = 67.7% vs. `val_acc` = 62.9%).

9 Conclusion and Future Work

This project shows that a structured progression from baseline tabular models to pre-trained transfer learning architectures leads to clear performance improvements. The top score of 56.4% reflects the real-world ambiguity of music genre definitions.

Future directions to improve performance include:

- Fine-tuning the upper layers of VGGish rather than keeping them entirely frozen.
- Using sequential models (LSTM or Transformers) over VGGish patch embeddings to preserve temporal progression.
- Combining audio representations with text metadata (artist, album, user tags) for multi-modal classification.